Greens function and the Poisson Integral formula on a Half-space

We use the method of reflection to compute the Green's function and the corresponding Poisson Kernel and Poisson integral formula for the case when the domain Ω is a half-space.

$$\Omega \equiv \{ (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n : x_n > 0 \}, \quad n \geq 2$$

and let $K(x)$ denote the fundamental solution given by

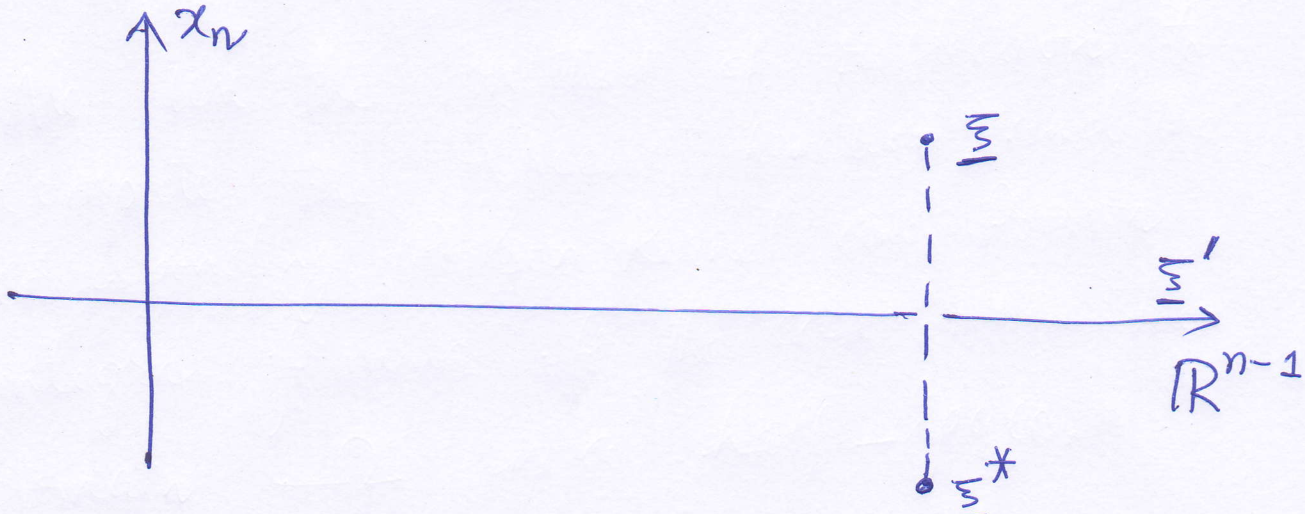
$$K(|x|) = \begin{cases} \frac{1}{2\pi} \log(|x|) \ , & n=2 \\ \frac{1}{(2-n)\omega_n} |x|^{2-n} \ , & n \geq 3. \end{cases}$$

<u>Notation</u>: $\underline{x} = (x_1, x_2, \ldots, x_n) = (\underline{x}', x_n)$

where $\underline{x}' \in \mathbb{R}^{n-1}$. Identify $(\underline{x}', 0) = \underline{x}'$.

For $\underline{\xi} = (\xi_1, \xi_2, \ldots, \xi_n) \in \Omega$,
define its reflection:

$$\underline{\xi}^* = (\xi_1, \xi_2, \ldots, \xi_{n-1}, -\xi_n).$$



Obviously $\underline{\xi}^* \notin \Omega$.

Notice that $K(|x-\xi^*|)$ is harmonic in $x \in \Omega$.

Moreover, for $\underline{x}' \in \mathbb{R}^{n-1}$, we have

$$|\underline{x}' - \underline{\xi}^*| = |\underline{x}' - \underline{\xi}|$$ and hence

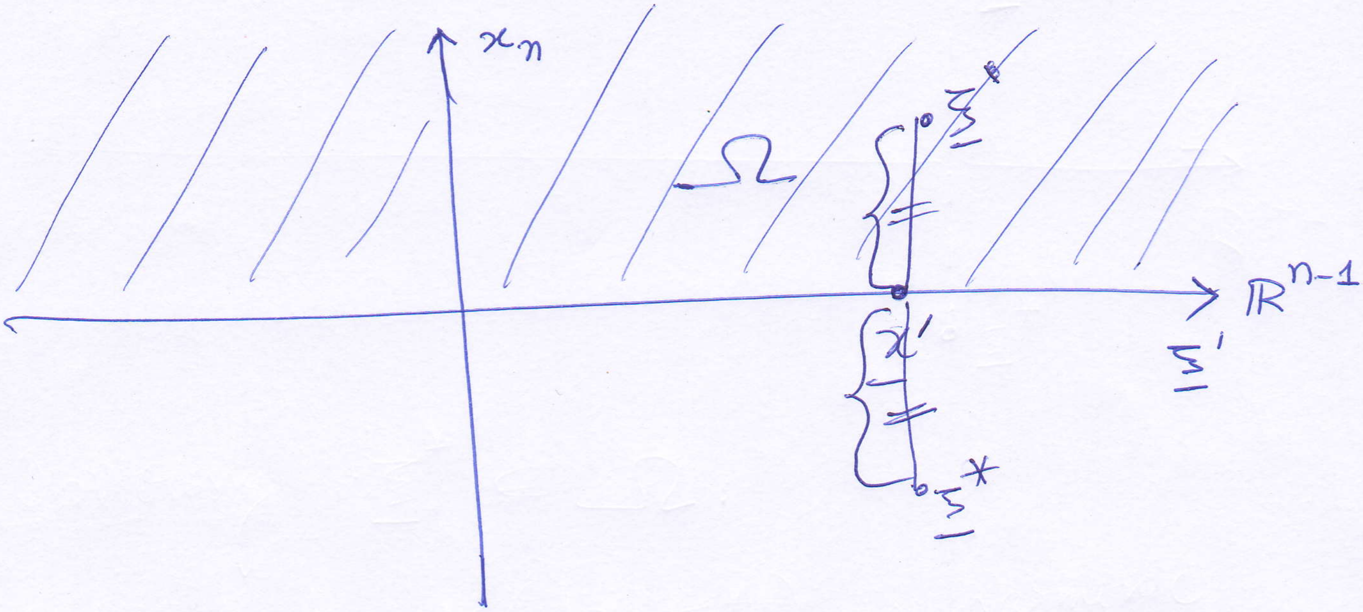
$$K(|\underline{x}' - \underline{\xi}^*|) = K(|\underline{x}' - \underline{\xi}|).$$

Now define the Green's function:

$$G(\underline{x}, \underline{\xi}) = K(|\underline{x}-\underline{\xi}|) - K(|\underline{x}-\underline{\xi}^*|)$$

Obviously $G(\underline{x}, \underline{\xi})$ is harmonic in Ω

and $G(\underline{x}, \underline{\xi}) = 0$ on the $\partial\Omega$.



What is $\partial\Omega$ (boundary of the Half-space)?

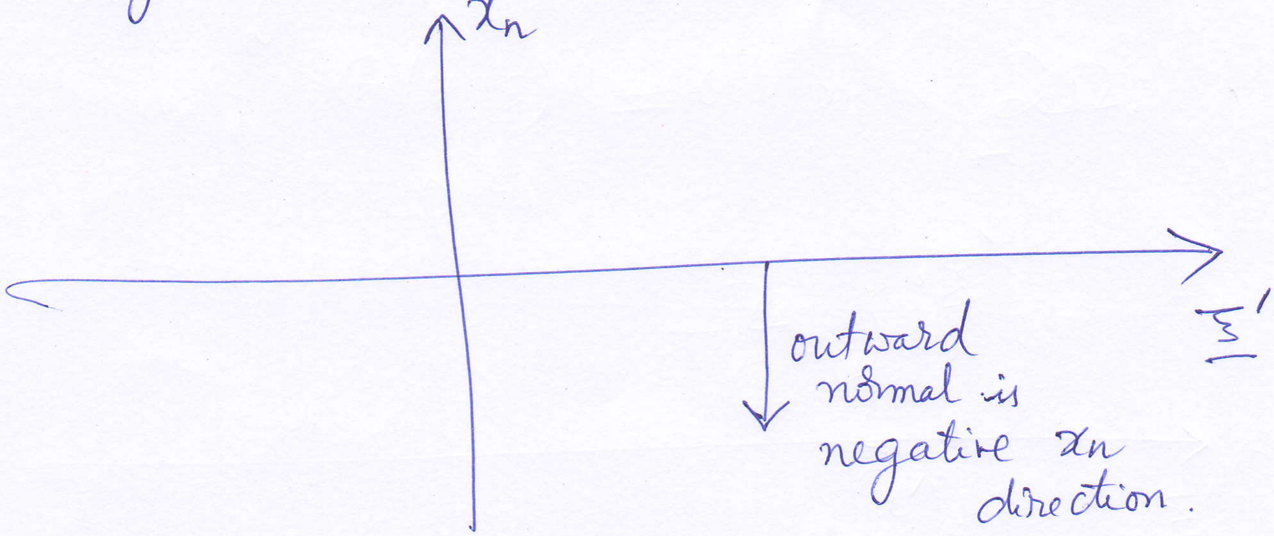
It is $\partial\Omega \equiv \{(x_1, x_2, \ldots, x_n) : x_n = 0\}$

ie $\underline{x}' \in \partial\Omega$. So

$$K(|\underline{x}'-\underline{\xi}|) = K(|\underline{x}'-\underline{\xi}^*|)$$

because $|\underline{x}'-\underline{\xi}| = |\underline{x}'-\underline{\xi}^*|$. (see figure).

To compute the Poisson Kernel, we must differentiate $G(\underline{x}, \underline{\xi})$ in the negative x_n direction



For $n \geq 2$, $\dfrac{\partial K(\underline{x}-\underline{\xi})}{\partial x_n}$

<u>$n=2$ Case</u> $\quad K(|\underline{x}-\underline{\xi}|) = \dfrac{1}{2\pi} \log(|\underline{x}-\underline{\xi}|)$

$$= \frac{1}{2\pi} \log\left(\sqrt{(x_1-\xi_1)^2+(x_2-\xi_2)^2}\right).$$

$$\frac{\partial K(|\underline{x}-\underline{\xi}|)}{\partial x_2} = \frac{(1/2\pi)}{\sqrt{(x_1-\xi_1)^2+(x_2-\xi_2)^2}} \cdot \frac{2(x_2-\xi_2)\cdot 1}{2\sqrt{(x_1-\xi_1)^2+(x_2-\xi_2)^2}}$$

$$= \frac{(x_2-\xi_2)}{\omega_2} |\underline{x}-\underline{\xi}|^{-2}.$$

Similarly for $n>2$,

$$\frac{\partial K(|\underline{x}-\underline{\xi}|)}{\partial x_n} = \frac{(x_n-\xi_n)}{\omega_n}|\underline{x}-\underline{\xi}|^{-n}.$$

Poisson Kernel $H(\underline{x},\underline{\xi}) = \left.\frac{-\partial G(\underline{x},\underline{\xi})}{\partial x_n}\right|_{x_n=0}$

$$= \left.\left[\frac{-\partial K(|\underline{x}-\underline{\xi}|)}{\partial x_n} + \frac{\partial K(|\underline{x}-\underline{\xi}^*|)}{\partial x_n}\right]\right|_{x_n=0}$$

$$= \left.\left[\frac{-(x_n-\xi_n)}{\omega_n}|\underline{x}-\underline{\xi}|^{-n} + \frac{(x_n+\xi_n)}{\omega_n}|\underline{x}-\underline{\xi}|^{-n}\right]\right|_{x_n=0}$$

$$= \frac{2\xi_n}{\omega_n}|\underline{x}'-\underline{\xi}|^{-n}, \text{ for } \underline{x}'\in\mathbb{R}^{n-1}.$$

Thus the solution of the Dirichlet problem:

$\Delta u = 0$ in $\Omega \equiv \{(x_1, x_2, \ldots x_n) : x_n>0\}$

$u=g$ on $\partial\Omega$ is

$$u(\underline{\xi}) = \frac{2\xi_n}{\omega_n}\int_{\mathbb{R}^{n-1}} \frac{g(\underline{x}')}{|\underline{x}'-\underline{\xi}|^n}\,d\underline{x}'.$$